

Phase 1

# Brainstorming & Ideation Phase

Problem discovery, ideas and proposed solution

## 1. Project Overview

**Project title:** AI StudyBuddy (tagline: *LearnMate - AI assistant for smarter learning and study support*).

AI StudyBuddy is an AI-powered learning assistance platform that simplifies the study process for students. Students upload study materials and receive concise summaries, flashcards, quizzes and personalized study plans generated through Generative AI. The backend is built with Node.js and Express.js using a RESTful architecture, MongoDB (via Mongoose) for storage, JWT authentication with Role-Based Access Control, and the Google Gemini AI API for natural language processing.

## 2. Problem Statement (Scenario-Based Case Study)

**Background:** Rahul is a second-year engineering student preparing for multiple semester examinations while also learning technical skills for internships and placements. His study materials are scattered across lecture notes, PDFs, textbooks, online articles and handwritten notes, making it difficult to organize and revise. Because of the large volume of content and limited time, he struggles to identify important concepts, create revision notes and evaluate his understanding before exams.

Traditional study methods need heavy manual effort to summarize lengthy material, prepare practice questions and create revision schedules. The process is repetitive, time-consuming and gives inconsistent learning outcomes.

### Pain points identified

- Difficulty understanding lengthy study materials within limited time.
- Manual preparation of notes consumes a significant amount of study time.
- Creating flashcards and revision questions is repetitive and labor-intensive.
- Lack of personalized study plans based on available time and examination schedules.
- No centralized platform to securely store AI-generated learning resources.
- Difficulty assessing knowledge through self-generated quizzes.
- Switching between multiple applications for note-taking, quizzes and scheduling reduces productivity.

## 3. Brainstormed Solution Ideas

| Problem                         | Idea generated                                                      |
|---------------------------------|---------------------------------------------------------------------|
| Lengthy materials, limited time | AI summary generation for faster revision                           |
| Repetitive flashcard creation   | AI flashcard generation for active recall                           |
| No way to self-assess           | AI multiple-choice quiz generation                                  |
| No personalized schedule        | AI study plan based on goal, hours per day and number of days       |
| Resources not stored centrally  | MongoDB storage with saved history per user                         |
| Data and access security        | JWT authentication, bcrypt password hashing, RBAC (Student / Admin) |
| Too many separate apps          | One centralized platform with a single set of REST APIs             |

## 4. Proposed Solution

A centralized, AI-powered backend that lets authenticated users upload or enter study material and use the Google Gemini AI API to generate educational resources automatically. The backend:

- Authenticates users using JWT-based security.
- Stores user information and study materials in MongoDB.
- Sends study content to the Google Gemini model through dedicated AI service modules.
- Generates concise summaries, flashcards, multiple-choice quizzes and personalized study plans.
- Stores generated learning resources for future retrieval and continuous learning.

## 5. Target Users

| User          | How they use AI StudyBuddy                                                                                                                                    |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Student       | Registers, logs in, uploads study materials and uses AI features (summary, flashcards, quiz, study plan). Can view and manage previously generated resources. |
| Administrator | Manages registered users and accounts, monitors application activity and AI service usage, reviews system logs and keeps backend operations running smoothly. |

## 6. Unique Value and Expected Outcome

- **Dynamic AI generation:** instead of predefined templates, the backend communicates with the Gemini model to generate high-quality educational content from the user's own material.
- **Reduced manual effort:** students focus on understanding concepts rather than organizing materials.
- **Modular and scalable:** authentication, business logic, database and AI processing are separate modules, so more AI features can be added with minimal architectural change.

**Outcome:** AI StudyBuddy serves as a scalable and intelligent educational platform that enhances productivity, promotes effective revision, and supports personalized learning for students preparing for academic examinations and competitive assessments.

## 7. Initial Technology Choices

| Layer               | Choice                                   |
|---------------------|------------------------------------------|
| Runtime / Framework | Node.js and Express.js (RESTful API)     |
| Database            | MongoDB with Mongoose ODM                |
| Security            | JWT, bcryptjs, Role-Based Access Control |
| AI Engine           | Google Gemini AI API                     |
| Frontend (client)   | React.js consuming the REST APIs         |